<u>Last time</u>: A metric $d$ satisfies...

- $d(x,y) \geq 0 \quad \forall x,y$
- $d(x,y) = 0 \iff x = y$
- $d(x,y) = d(y,x) \quad \forall x,y$
- $d(x,z) \leq d(x,y) + d(y,z) \quad \forall x,y,z$

<u>Ex</u> On $\mathbb{R}^2$,

① $d_1(x,y) = |x_1 - y_1| + |x_2 - y_2|$

② $d_2(x,y) = \sqrt{|x_1-y_1|^2 + |x_2-y_2|^2}$

③ $d_\infty(x,y) = \max\{|x_1-y_1|, |x_2-y_2|\}$

- Why do ①–③ satisfy the triangle inequality?

<u>Prop</u> (Hölder's inequality) Fix $1 \leq p \leq \infty$, and let $1 \leq q \leq \infty$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$. For any $(x_1, \dots, x_n)$, $(y_1, \dots, y_n) \in \mathbb{R}^n$ we have

$$\sum_{k=1}^n |x_k y_k| \leq \left(\sum_{k=1}^n |x_k|^p\right)^{\frac{1}{p}} \left(\sum_{k=1}^n |y_k|^q\right)^{\frac{1}{q}}.$$

When $p = \infty$, replace $\left(\sum_{k=1}^n |x_k|^p\right)^{\frac{1}{p}}$ by $\max_{k=1,\dots,n} |x_k|$ above.

<u>Rmk</u> When $p = 2$ ($\Rightarrow q = 2$), we call this the Cauchy–Schwarz inequality.

<u>Pf</u>: Case: $p = 1$. Then $q = \infty$ satisfies $\frac{1}{p} + \frac{1}{q} = 1 + 0 = 1$.

$$\sum_{k=1}^{n} |x_k y_k| = \sum_{k=1}^{n} |x_k| \cdot |y_k| \le \sum_{k=1}^{n} |x_k| \cdot \left(\max_{\ell=1,\dots,n} |y_\ell|\right)$$

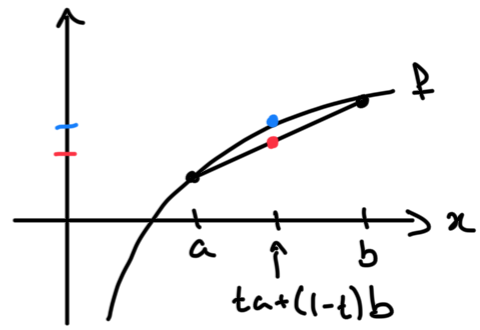
$$= \left(\sum_{k=1}^{n} |x_k|\right)\left(\max_{\ell=1,\dots,n} |y_\ell|\right).$$

Case: $p=\infty$. Then $q=1$. Swap $x_k$ and $y_k$ above.

Case: $1<p<\infty$. We will use the fact that $f:(0,\infty)\to\mathbb{R}$, $f(x)=\log x$ is a concave function:

$\forall a,b\in(0,\infty)$, $\forall t\in(0,1)$,

$$\underline{t\,f(a)+(1-t)\,f(b)} \le \underline{f(ta+(1-t)b)}$$




$$\Rightarrow \underbrace{t\log a+(1-t)\log b}_{\log(a^t)+\log(b^{1-t}) = \log(a^t b^{1-t})} \le \log(ta+(1-t)b)$$

$$\Rightarrow \boxed{a^t b^{1-t} \le ta+(1-t)b}$$

We apply this to: $a=\dfrac{|x_k|^p}{\sum_{\ell=1}^{n}|x_\ell|^p}$, $b=\dfrac{|y_k|^q}{\sum_{\ell=1}^{n}|y_\ell|^q}$

$$t=\frac{1}{p} \quad\Rightarrow\quad 1-t=1-\frac{1}{p}=\frac{1}{q}$$

$$\Rightarrow \frac{|x_k|}{\left(\sum_{\ell=1}^{n}|x_\ell|^p\right)^{\frac{1}{p}}}\cdot\frac{|y_k|}{\left(\sum_{\ell=1}^{n}|y_\ell|^q\right)^{\frac{1}{q}}} \le \frac{1}{p}\,\frac{|x_k|^p}{\sum_{\ell=1}^{n}|x_\ell|^p}+\frac{1}{q}\,\frac{|y_k|^q}{\sum_{\ell=1}^{n}|y_\ell|^q}$$

Sum over $k=1,2,\dots,n$:

$$\Longrightarrow \sum_{k=1}^{n} \frac{|x_k|\cdot|y_k|}{\left(\sum_{\ell=1}^{n}|x_\ell|^p\right)^{\frac{1}{p}}\left(\sum_{\ell=1}^{n}|y_\ell|^q\right)^{\frac{1}{q}}} \le \frac{1}{p}\sum_{k=1}^{n}\frac{|x_k|^p}{\sum_{\ell=1}^{n}|x_\ell|^p} + \frac{1}{q}\sum_{k=1}^{n}\frac{|y_k|^q}{\sum_{\ell=1}^{n}|y_\ell|^q}$$

$$= \frac{1}{p}+\frac{1}{q} = 1$$

$$\Longrightarrow \sum_{k=1}^{n}|x_k y_k| \le \left(\sum_{\ell=1}^{n}|x_\ell|^p\right)^{\frac{1}{p}}\left(\sum_{\ell=1}^{n}|y_\ell|^q\right)^{\frac{1}{q}}. \qquad \square$$

<u>Cor</u> (Minkowski's inequality) For any $1\le p\le\infty$ and $(x_1,\dots,x_n),(y_1,\dots,y_n)\in\mathbb{R}^n$, we have

$$\left(\sum_{k=1}^{n}|x_k+y_k|^p\right)^{\frac{1}{p}} \le \left(\sum_{k=1}^{n}|x_k|^p\right)^{\frac{1}{p}} + \left(\sum_{k=1}^{n}|y_k|^p\right)^{\frac{1}{p}}.$$

<u>Pf</u>: Case: $p=1$.

$$\sum_{k=1}^{n}|x_k+y_k| \le \sum_{k=1}^{n}\left(|x_k|+|y_k|\right) = \sum_{k=1}^{n}|x_k| + \sum_{k=1}^{n}|y_k|.$$

Case: $p=\infty$.

$$\max_{k=1,\dots,n}|x_k+y_k| \le \max_{k=1,\dots,n}|x_k| + \max_{k=1,\dots,n}|y_k|$$

Case: $1<p<\infty$.

$$\sum_{k=1}^{n}|x_k+y_k|^p = \sum_{k=1}^{n}\underbrace{|x_k+y_k|}_{\le |x_k|+|y_k|}\cdot|x_k+y_k|^{p-1}$$

$$\le \sum_{k=1}^{n}|x_k|\cdot|x_k+y_k|^{p-1} + \sum_{k=1}^{n}|y_k|\cdot|x_k+y_k|^{p-1}$$

$$\overset{\text{Hölder}}{\leq} \left(\sum_{k=1}^{n} |x_k|^p\right)^{\frac{1}{p}} \left(\sum_{k=1}^{n} |x_k+y_k|^{(p-1)q}\right)^{\frac{1}{q}}$$

$$+ \left(\sum_{k=1}^{n} |y_k|^p\right)^{\frac{1}{p}} \left(\sum_{k=1}^{n} |x_k+y_k|^{(p-1)q}\right)^{\frac{1}{q}}$$

where

$$\frac{1}{p} + \frac{1}{q} = 1 \quad \Rightarrow \quad \frac{1}{q} = 1 - \frac{1}{p} = \frac{p-1}{p} \quad \Rightarrow \quad q = \frac{p}{p-1}$$

Get:

$$\sum_{k=1}^{n} |x_k+y_k|^p \leq \left[\left(\sum_{k=1}^{n} |x_k|^p\right)^{\frac{1}{p}} + \left(\sum_{k=1}^{n} |y_k|^p\right)^{\frac{1}{p}}\right]\left(\sum_{k=1}^{n} |x_k+y_k|^p\right)^{1-\frac{1}{p}}$$

$$\Rightarrow \left(\sum_{k=1}^{n} |x_k+y_k|^p\right)^{\frac{1}{p}} \leq \left(\sum_{k=1}^{n} |x_k|^p\right)^{\frac{1}{p}} + \left(\sum_{k=1}^{n} |y_k|^p\right)^{\frac{1}{p}}. \quad \square$$

<u>Cor</u> For any $1 \leq p \leq \infty$, the function $d_p : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$,

$$d_p(x,y) = \begin{cases} \left(\sum_{k=1}^{n} |x_k - y_k|^p\right)^{\frac{1}{p}} & \text{if } 1 \leq p \leq \infty \\ \max_{k=1,\dots,n} |x_k - y_k| & \text{if } p = \infty \end{cases}$$

is a metric on $\mathbb{R}^n$.

<u>Pf</u>: We'll verify the triangle inequality. (The other 3 properties are easy: exercise.)

By Minkowski's inequality,

$$\left(\sum_{k=1}^{n} |x_k - z_k|^p\right)^{\frac{1}{p}} \leq \left(\sum_{k=1}^{n} |x_k - y_k|^p\right)^{\frac{1}{p}} + \left(\sum_{k=1}^{n} |y_k - z_k|^p\right)^{\frac{1}{p}}$$

$$= x_k - y_k + y_k - z_k$$

$$\Rightarrow \quad d_p(x,z) \leq d_p(x,y) + d_p(y,z). \qquad \square$$

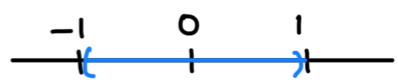
<u>Def</u> Let $(X, d)$ be a metric space. The <u>open ball</u> (or <u>neighborhood</u>) of radius $r>0$ centered at $a \in X$ is

$$B_r(a) = \{x \in X : d(a,x) < r\}.$$

<u>Ex</u> $X = \mathbb{R}$ with the metric...

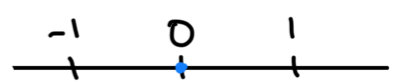
① $d(x,y) = |x-y|$. Then

$$B_1(0) = \{x \in \mathbb{R} : |x-0| < 1\}$$
$$= (-1, 1)$$




② $d$ = discrete metric. Then

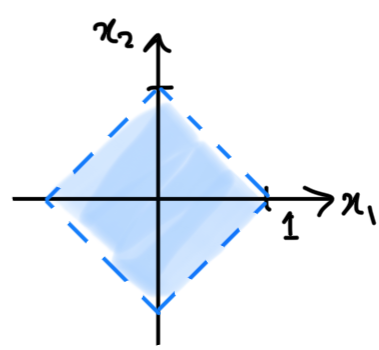
$$B_1(0) = \{x \in \mathbb{R} : \underbrace{d(x,0)}_{= \begin{cases} 0 & x=0 \\ 1 & x \neq 0 \end{cases}} < 1\}$$
$$= \{0\}$$




<u>Ex</u> $\mathbb{R}^2$ with the metric...

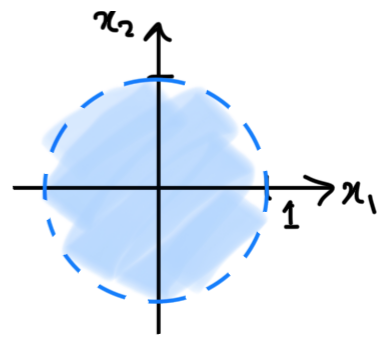
① $d_1$. Then

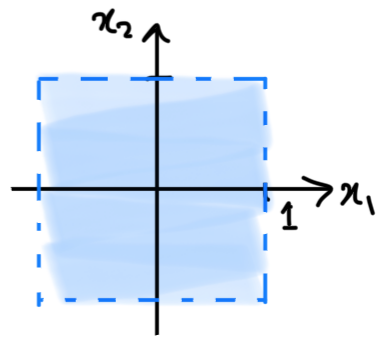
$$B_1(0) = \{(x_1, x_2) \in \mathbb{R}^2 : |x_1 - 0| + |x_2 - 0| < 1\}$$




② $d_2$. Then

$$B_1(0) = \{ (x_1, x_2) \in \mathbb{R}^2 : \sqrt{|x_1|^2 + |x_2|^2} < 1 \}$$




③ $d_\infty$. Then

$$B_1(0) = \{ (x_1, x_2) \in \mathbb{R}^2 : \max\{|x_1|, |x_2|\} < 1 \}$$